PETGEM CI Summary Report

Compile PETGEM Log

[CC] src/csem_kernel.c [CC] src/common.c [CC] src/inputs.c [CC] src/transmitter.c [CC] src/grid.c [CC] src/assembly.c [CC] src/hvfem.c [CC] src/solver.c [CC] src/postprocessing.c [LD] build/csem_kernel

Documentation Log

> > > [CLEAN] Cleaning documentation rm -rf docs/build/html/* docs/doxygen/* docs/source/api/* docs/source/readme/* docs/build/html [DOC] Generating Doxygen XML doxygen Doxyfile warning: ignoring unsupported tag 'CREATE_SUBDIRS_LEVEL' at line 93, file Doxyfile warning: ignoring unsupported tag 'MARKDOWN_ID_STYLE' at line 375, file Doxyfile warning: ignoring unsupported tag 'TIMESTAMP' at line 507, file Doxyfile warning: ignoring unsupported tag 'SHOW_HEADERFILE' at line 652, file Doxyfile warning: ignoring unsupported tag 'WARN_IF_INCOMPLETE_DOC' at line 870, file Doxyfile warning: ignoring unsupported tag 'WARN_IF_UNDOC_ENUM_VAL' at line 888, file Doxyfile warning: ignoring unsupported tag 'WARN_LINE_FORMAT' at line 925, file Doxyfile warning: ignoring unsupported tag 'INPUT_FILE_ENCODING' at line 967, file Doxyfile warning: ignoring unsupported tag 'FORTRAN_COMMENT_AFTER' at line 1168, file Doxyfile warning: ignoring unsupported tag 'HTML_COLORSTYLE' at line 1421, file Doxyfile warning: ignoring unsupported tag 'HTML_CODE_FOLDING' at line 1477, file Doxyfile warning: ignoring unsupported tag 'DOCSET_FEEDURL' at line 1520, file Doxyfile warning: ignoring unsupported tag 'SITEMAP_URL' at line 1617, file Doxyfile warning: ignoring unsupported tag 'FULL_SIDEBAR' at line 1741, file Doxyfile warning: ignoring unsupported tag 'OBFUSCATE_EMAILS' at line 1772, file Doxyfile warning: ignoring unsupported tag 'MATHJAX_VERSION' at line 1820, file Doxyfile warning: ignoring unsupported tag 'GENERATE_SQLITE3' at line 2317, file Doxyfile warning: ignoring unsupported tag 'SQLITE3_OUTPUT' at line 2325, file Doxyfile warning: ignoring unsupported tag 'SQLITE3_RECREATE_DB' at line 2333, file Doxyfile warning: ignoring unsupported tag 'DOT_COMMON_ATTR' at line 2538, file Doxyfile warning: ignoring unsupported tag 'DOT_EDGE_ATTR' at line 2547, file Doxyfile warning: ignoring unsupported tag

'DOT_NODE_ATTR' at line 2555, file Doxyfile warning: ignoring unsupported tag 'DIR_GRAPH_MAX_DEPTH' at line 2721, file Doxyfile warning: ignoring unsupported tag 'MSCGEN_TOOL' at line 2855, file Doxyfile Doxygen version used: 1.9.1 Notice: Output directory './docs/doxygen' does not exist. I have created it for you. Searching for include files… Searching for example files… Searching for images… Searching for dot files… Searching for msc files… Searching for dia files… Searching for files to exclude Searching INPUT for files to process… Searching for files in directory /app/src Searching for files in directory /app/include Reading and parsing tag files Parsing files Preprocessing /app/src/assembly.c… Parsing file /app/src/assembly.c… Preprocessing /app/src/common.c… Parsing file /app/src/common.c… Preprocessing /app/src/csem_kernel.c… Parsing file /app/src/csem_kernel.c… Preprocessing /app/src/grid.c… Parsing file /app/src/grid.c… Preprocessing /app/src/hvfem.c… Parsing file /app/src/hvfem.c… Preprocessing /app/src/inputs.c… Parsing file /app/src/inputs.c… Preprocessing /app/src/postprocessing.c… Parsing file /app/src/postprocessing.c… Preprocessing /app/src/solver.c… Parsing file /app/src/solver.c… Preprocessing /app/src/transmitter.c… Parsing file /app/src/transmitter.c… Preprocessing /app/include/assembly.h… Parsing file /app/include/assembly.h… Preprocessing /app/include/common.h… Parsing file /app/include/common.h… Preprocessing /app/include/constants.h… Parsing file /app/include/constants.h… Preprocessing /app/include/grid.h… Parsing file /app/include/grid.h… Preprocessing /app/include/hvfem.h… Parsing file /app/include/hvfem.h… Preprocessing /app/include/inputs.h… Parsing file /app/include/inputs.h… Preprocessing /app/include/postprocessing.h… Parsing file /app/include/postprocessing.h… Preprocessing /app/include/receivers.h… Parsing file /app/include/receivers.h… Preprocessing /app/include/solver.h… Parsing file /app/include/solver.h… Preprocessing /app/include/transmitter.h… Parsing file /app/include/transmitter.h… Preprocessing /app/include/version.h… Parsing file /app/include/version.h… Building macro definition list… Building group list… Building directory list… Building namespace list… Building file list… Building class list… Computing nesting relations for classes… Associating documentation with classes… Building example list… Searching for enumerations… Searching for documented typedefs… Searching for members imported via using declarations… Searching for included using directives… Searching for documented variables… Building interface member list… Building member list… Searching for friends… Searching for documented defines… Computing class inheritance relations… Computing class usage relations… Flushing cached template relations that have become invalid… Computing class relations… Add enum values to enums… Searching for member function documentation… Creating members for template instances… Building page list… Search for main page… Computing page relations… Determining the scope of groups… Sorting lists… Determining which enums are documented Computing member relations… Building full member lists recursively… Adding members to member groups. Computing member references… Inheriting documentation… Generating disk names… Adding source references… Adding xrefitems… Sorting member lists… Setting anonymous enum type… Computing dependencies between directories… Generating citations page… Counting members… Counting data structures… Resolving user defined references… Finding anchors and sections in the documentation… Transferring function references… Combining using relations… Adding members to index pages… Correcting members for VHDL… Computing tooltip texts… Generating style sheet… Generating search indices… Generating example documentation… Generating file sources… Generating code for file include/assembly.h… Generating code for file include/common.h… Generating code for file include/constants.h… Generating code for file include/grid.h… Generating code for file include/hvfem.h… Generating code for file include/inputs.h… Generating code for file include/postprocessing.h… Generating code for file include/receivers.h… Generating code for file include/solver.h… Generating code for file include/transmitter.h… Generating code for file include/version.h… Generating file documentation… Generating docs for file include/assembly.h… Generating docs for file include/common.h… Generating docs for file include/constants.h… Generating docs for file include/grid.h… Generating docs for file include/hvfem.h… Generating docs for file include/inputs.h… Generating docs for file include/postprocessing.h… Generating docs for file include/receivers.h… Generating docs for file include/solver.h… Generating docs for file include/transmitter.h… Generating docs for file include/version.h… Generating docs for file src/assembly.c… Generating docs for file src/common.c… Generating docs for file src/csem_kernel.c… Generating docs for file src/grid.c… Generating docs for file src/hvfem.c… Generating docs for file src/inputs.c… Generating docs for file src/postprocessing.c… Generating docs for file src/solver.c… Generating docs for file src/transmitter.c… Generating page documentation… Generating group documentation… Generating class documentation… Generating docs for compound CsemSource… Generating docs for compound

CsemSourceSet… Generating docs for compound Grid… Generating docs for compound Params… Generating namespace index… Generating graph info page… Generating directory documentation… Generating dependency graph for directory src finalizing index lists… writing tag file… Generating XML output… Generating XML output for class CsemSource Generating XML output for class CsemSourceSet Generating XML output for class Grid Generating XML output for class Params Generating XML output for file assembly.h Generating XML output for file common.h Generating XML output for file constants.h Generating XML output for file grid.h Generating XML output for file hvfem.h Generating XML output for file inputs.h Generating XML output for file postprocessing.h Generating XML output for file receivers.h Generating XML output for file solver.h Generating XML output for file transmitter.h Generating XML output for file version.h Generating XML output for file assembly.c Generating XML output for file common.c Generating XML output for file csem_kernel.c Generating XML output for file grid.c Generating XML output for file hvfem.c Generating XML output for file inputs.c Generating XML output for file postprocessing.c Generating XML output for file solver.c Generating XML output for file transmitter.c Generate XML output for dir /app/include/ Generate XML output for dir /app/src/ Running plantuml with JAVA… Running dot… lookup cache used 141/65536 hits=1043 misses=146 finished… [DOC] Running API generator (api_generator) python3 scripts/auto_doc/api_rst_generator.py Directory created: ../../docs/source/api Generated: ../../docs/source/api/grid.rst Generated: ../../docs/source/api/inputs.rst Generated: ../../docs/source/api/solver.rst Generated: ../../docs/source/api/transmitter.rst Generated: ../../docs/source/api/hvfem.rst Generated: ../../docs/source/api/csem_kernel.rst Generated: ../../docs/source/api/postprocessing.rst Generated: ../../docs/source/api/assembly.rst Generated: ../../docs/source/api/common.rst Generated/Updated API index: ../../docs/source/api/index.rst [DOC] Building Sphinx HTML documentation… LC_ALL=C.UTF-8 LANG=C.UTF-8 python3 -m sphinx -b html docs/source docs/build/html Running Sphinx v4.3.2 making output directory… done building [mo]: targets for 0 po files that are out of date building [html]: targets for 17 source files that are out of date updating environment: [new config] 17 added, 0 changed, 0 removed reading sources… [5%] api/assembly reading sources… [11%] api/common reading sources… [17%] api/csem_kernel reading sources… [23%] api/grid reading sources… [29%] api/hvfem reading sources… [35%] api/index reading sources… [41%] api/inputs reading sources… [47%] api/postprocessing reading sources… [52%] api/solver reading sources… [58%] api/transmitter

> > > reading sources… [64%] index reading sources… [70%] manual/about_petgem reading sources… [76%] manual/contact reading sources… [82%] manual/contributing reading sources… [88%] manual/guide reading sources… [94%] manual/install reading sources… [100%] manual/publications

looking for now-outdated files… none found pickling environment… done checking consistency… done preparing documents… done writing output… [5%] api/assembly writing output… [11%] api/common writing output… [17%] api/csem_kernel writing output… [23%] api/grid writing output… [29%] api/hvfem writing output… [35%] api/index writing output… [41%] api/inputs writing output… [47%] api/postprocessing writing output… [52%] api/solver writing output… [58%] api/transmitter writing output… [64%] index writing output… [70%] manual/about_petgem writing output… [76%] manual/contact writing output… [82%] manual/contributing writing output… [88%] manual/guide writing output… [94%] manual/install writing output… [100%] manual/publications

generating indices… genindex done writing additional pages… search done copying images… [100%] _static/petgem_logo.png

copying static files… done copying extra files… done dumping search index in English (code: en)… done dumping object inventory… done build succeeded.

The HTML pages are in docs/build/html. >>> [DOCS] HTML documentation generated in docs/build/html Total functions: 75 Documented functions: 75 Documentation Coverage: 100.00% Generated badge: ./docs/doc_coverage_badge.svg

Test Log

Info : Running 'gmsh tests/canonical_model/mesh.geo -3' [Gmsh 4.14.1, 1 node, max. 1 thread] Info : Started on Fri Sep 12 13:16:25 2025 Info : Reading 'tests/canonical_model/mesh.geo'… Info : Done reading 'tests/canonical_model/mesh.geo' Info : Meshing 1D… Info : [0%] Meshing curve 1 (Line) Info : [10%] Meshing curve 2 (Line) Info : [10%] Meshing curve 3 (Line) Info : [10%] Meshing curve 4 (Line) Info : [20%] Meshing curve 5 (Line) Info : [20%] Meshing curve 6 (Line) Info : [20%] Meshing curve 7 (Line) Info : [20%] Meshing curve 8 (Line) Info : [30%] Meshing curve 9 (Line) Info : [30%] Meshing curve 10 (Line) Info : [30%] Meshing curve 11 (Line) Info : [40%] Meshing curve 12 (Line) Info : [40%] Meshing curve 13 (Line) Info : [40%] Meshing curve 14 (Line) Info : [40%] Meshing curve 15 (Line) Info : [50%] Meshing curve 16 (Line) Info : [50%] Meshing curve 17 (Line) Info : [50%] Meshing curve 18 (Line) Info : [60%] Meshing curve 19 (Line) Info : [60%] Meshing curve 20 (Line) Info : [60%] Meshing curve 21 (Line) Info : [60%] Meshing curve 22 (Line) Info : [70%] Meshing curve 23 (Line) Info : [70%] Meshing curve 24 (Line) Info : [70%] Meshing curve 25 (Line) Info : [70%] Meshing curve 26 (Line) Info : [80%] Meshing curve 27 (Line) Info : [80%] Meshing curve 28 (Line) Info : [80%] Meshing curve 29 (Line) Info : [90%] Meshing curve 30 (Line) Info : [90%] Meshing curve 31 (Line) Info : [90%] Meshing curve 32 (Line) Info : [90%] Meshing curve 34 (Line) Info : [100%] Meshing curve 35 (Line) Info : [100%] Meshing curve 36 (Line) Info : [100%] Meshing curve 37 (Line) Info : Done meshing 1D (Wall 0.00686555s, CPU 0.008885s) Info : Meshing 2D… Info : [0%] Meshing surface 1 (Plane, Frontal-Delaunay) Info : [10%]

Meshing surface 2 (Plane, Frontal-Delaunay) Info : [10%] Meshing surface 3 (Plane, Frontal-Delaunay) Info : [20%] Meshing surface 4 (Plane, Frontal-Delaunay) Info : [20%] Meshing surface 5 (Plane, Frontal-Delaunay) Info : [30%] Meshing surface 6 (Plane, Frontal-Delaunay) Info : [30%] Meshing surface 7 (Plane, Frontal-Delaunay) Info : [40%] Meshing surface 8 (Plane, Frontal-Delaunay) Info : [40%] Meshing surface 9 (Plane, Frontal-Delaunay) Info : [50%] Meshing surface 10 (Plane, Frontal-Delaunay) Info : [50%] Meshing surface 11 (Plane, Frontal-Delaunay) Info : [60%] Meshing surface 12 (Plane, Frontal-Delaunay) Info : [60%] Meshing surface 13 (Plane, Frontal-Delaunay) Info : [70%] Meshing surface 14 (Plane, Frontal-Delaunay) Info : [70%] Meshing surface 15 (Plane, Frontal-Delaunay) Info : [80%] Meshing surface 16 (Plane, Frontal-Delaunay) Info : [80%] Meshing surface 17 (Plane, Frontal-Delaunay) Info : [90%] Meshing surface 18 (Plane, Frontal-Delaunay) Info : [90%] Meshing surface 19 (Plane, Frontal-Delaunay) Info : [100%] Meshing surface 20 (Plane, Frontal-Delaunay) Info : [100%] Meshing surface 21 (Plane, Frontal-Delaunay) Info : Done meshing 2D (Wall 0.100257s, CPU 0.129851s) Info : Meshing 3D... Info : 3D Meshing 4 volumes with 1 connected component Info : Tetrahedrizing 3030 nodes... Info : Done tetrahedrizing 3038 nodes (Wall 0.0383661s, CPU 0.03516s) Info : Reconstructing mesh... Info : - Creating surface mesh Info : - Identifying boundary edges Info : - Recovering boundary Info : - Added 8 Steiner points Info : Done reconstructing mesh (Wall 0.0888441s, CPU 0.07809s) Info : Found volume 1 Info : Found volume 4 Info : Found volume 3 Info : Found volume 2 Info : It. 0 - 0 nodes created - worst tet radius 6.5706 (nodes removed 0 0) Info : It. 500 - 494 nodes created - worst tet radius 2.41349 (nodes removed 0 6) Info : It. 1000 - 993 nodes created - worst tet radius 1.96959 (nodes removed 0 7) Info : It. 1500 - 1493 nodes created - worst tet radius 1.72506 (nodes removed 0 7) Info : It. 2000 - 1993 nodes created - worst tet radius 1.58124 (nodes removed 0 7) Info : It. 2500 - 2493 nodes created - worst tet radius 1.76683 (nodes removed 0 7) Info : It. 3000 - 2993 nodes created - worst tet radius 1.39458 (nodes removed 0 7) Info : It. 3500 - 3485 nodes created - worst tet radius 1.33269 (nodes removed 0 15) Info : It. 4000 - 3985 nodes created - worst tet radius 1.28202 (nodes removed 0 15) Info : It. 4500 - 4485 nodes created - worst tet radius 1.23473 (nodes removed 0 15) Info : It. 5000 - 4985 nodes created - worst tet radius 1.19299 (nodes removed 0 15) Info : It. 5500 - 5483 nodes created - worst tet radius 1.15483 (nodes removed 0 17) Info : It. 6000 - 5982 nodes created - worst tet radius 1.12367 (nodes removed 0 18) Info : It. 6500 - 6481 nodes created - worst tet radius 1.09755 (nodes removed 0 19) Info : It. 7000 - 6981 nodes created - worst tet radius 1.07116 (nodes removed 0 19) Info : It. 7500 - 7481 nodes created - worst tet radius 1.04876 (nodes removed 0 19) Info : It. 8000 - 7981 nodes created - worst tet radius 1.02801 (nodes removed 0 19) Info : It. 8500 - 8481 nodes created - worst tet radius 1.00789 (nodes removed 0 19) Info : 3D refinement terminated (11739 nodes total): Info : - 3 Delaunay cavities modified for star shapeness Info : - 19 nodes could not be inserted Info : - 69767 tetrahedra created in 0.408712 sec. (170699 tets/s) Info : 18 node relocations Info : Done meshing 3D (Wall 0.726546s, CPU 0.709373s) Info : Optimizing mesh... Info : Optimizing volume 1 Info : Optimization starts (volume = 2.695e+10) with worst = 0.0126867 / average = 0.76209: Info : 0.00 < quality < 0.10 : 15 elements Info : 0.10 < quality < 0.20 : 25 elements Info : 0.20 < quality < 0.30 : 37 elements Info : 0.30 < quality < 0.40 : 58 elements Info : 0.40 < quality < 0.50 : 81 elements Info : 0.50 < quality < 0.60 : 150 elements Info : 0.60 < quality < 0.70 : 551 elements Info : 0.70 < quality < 0.80 : 826 elements Info : 0.80 < quality < 0.90 : 1062 elements Info : 0.90 <

quality < 1.00 : 555 elements Info : 76 edge swaps, 3 node relocations (volume = 2.695e+10): worst = 0.258139 / average = 0.776045 (Wall 0.00139255s, CPU 0.001392s) Info : 78 edge swaps, 3 node relocations (volume = 2.695e+10): worst = 0.300235 / average = 0.776415 (Wall 0.00173741s, CPU 0.001737s) Info : No ill-shaped tets in the mesh :-) Info : 0.00 < quality < 0.10 : 0 elements Info : 0.10 < quality < 0.20 : 0 elements Info : 0.20 < quality < 0.30 : 0 elements Info : 0.30 < quality < 0.40 : 62 elements Info : 0.40 < quality < 0.50 : 79 elements Info : 0.50 < quality < 0.60 : 147 elements Info : 0.60 < quality < 0.70 : 544 elements Info : 0.70 < quality < 0.80 : 830 elements Info : 0.80 < quality < 0.90 : 1072 elements Info : 0.90 < quality < 1.00 : 558 elements Info : Optimizing volume 2 Info : Optimization starts (volume = 1.925e+09) with worst = 0.145936 / average = 0.52348: Info : 0.00 < quality < 0.10 : 0 elements Info : 0.10 < quality < 0.20 : 12 elements Info : 0.20 < quality < 0.30 : 6 elements Info : 0.30 < quality < 0.40 : 28 elements Info : 0.40 < quality < 0.50 : 56 elements Info : 0.50 < quality < 0.60 : 1089 elements Info : 0.60 < quality < 0.70 : 2 elements Info : 0.70 < quality < 0.80 : 0 elements Info : 0.80 < quality < 0.90 : 0 elements Info : 0.90 < quality < 1.00 : 0 elements Info : 1 edge swaps, 5 node relocations (volume = 1.925e+09): worst = 0.145936 / average = 0.523745 (Wall 0.000377988s, CPU 0.000377s) Info : No ill-shaped tets in the mesh :-) Info : 0.00 < quality < 0.10 : 0 elements Info : 0.10 < quality < 0.20 : 12 elements Info : 0.20 < quality < 0.30 : 4 elements Info : 0.30 < quality < 0.40 : 28 elements Info : 0.40 < quality < 0.50 : 59 elements Info : 0.50 < quality < 0.60 : 1089 elements Info : 0.60 < quality < 0.70 : 2 elements Info : 0.70 < quality < 0.80 : 0 elements Info : 0.80 < quality < 0.90 : 0 elements Info : 0.90 < quality < 1.00 : 0 elements Info : Optimizing volume 3 Info : Optimization starts (volume = 1.925e+10) with worst = 0.0153404 / average = 0.754145: Info : 0.00 < quality < 0.10 : 58 elements Info : 0.10 < quality < 0.20 : 200 elements Info : 0.20 < quality < 0.30 : 435 elements Info : 0.30 < quality < 0.40 : 718 elements Info : 0.40 < quality < 0.50 : 1223 elements Info : 0.50 < quality < 0.60 : 2165 elements Info : 0.60 < quality < 0.70 : 4188 elements Info : 0.70 < quality < 0.80 : 7209 elements Info : 0.80 < quality < 0.90 : 9877 elements Info : 0.90 < quality < 1.00 : 5157 elements Info : 665 edge swaps, 32 node relocations (volume = 1.925e+10): worst = 0.208316 / average = 0.766761 (Wall 0.0131225s, CPU 0.01306s) Info : 672 edge swaps, 34 node relocations (volume = 1.925e+10): worst = 0.265054 / average = 0.76689 (Wall 0.0160241s, CPU 0.015962s) Info : No ill-shaped tets in the mesh :-) Info : 0.00 < quality < 0.10 : 0 elements Info : 0.10 < quality < 0.20 : 0 elements Info : 0.20 < quality < 0.30 : 2 elements Info : 0.30 < quality < 0.40 : 689 elements Info : 0.40 < quality < 0.50 : 1236 elements Info : 0.50 < quality < 0.60 : 2143 elements Info : 0.60 < quality < 0.70 : 4214 elements Info : 0.70 < quality < 0.80 : 7273 elements Info : 0.80 < quality < 0.90 : 9949 elements Info : 0.90 < quality < 1.00 : 5136 elements Info : Optimizing volume 4 Info : Optimization starts (volume = 1.925e+10) with worst = 0.00510254 / average = 0.755429: Info : 0.00 < quality < 0.10 : 89 elements Info : 0.10 < quality < 0.20 : 225 elements Info : 0.20 < quality < 0.30 : 471 elements Info : 0.30 < quality < 0.40 : 760 elements Info : 0.40 < quality < 0.50 : 1301 elements Info : 0.50 < quality < 0.60 : 2302 elements Info : 0.60 < quality < 0.70 : 4396 elements Info : 0.70 < quality < 0.80 : 8026 elements Info : 0.80 < quality < 0.90 : 10775 elements Info : 0.90 < quality < 1.00 : 5639 elements Info : 748 edge swaps, 21 node relocations (volume = 1.925e+10): worst = 0.211918 / average = 0.7679 (Wall 0.0146659s, CPU 0.014665s) Info : 757 edge swaps, 23 node relocations (volume = 1.925e+10): worst = 0.211918 / average = 0.767986 (Wall

0.0179345s, CPU 0.017921s) Info : No ill-shaped tets in the mesh :-) Info : 0.00 < quality < 0.10 : 0 elements Info : 0.10 < quality < 0.20 : 0 elements Info : 0.20 < quality < 0.30 : 7 elements Info : 0.30 < quality < 0.40 : 764 elements Info : 0.40 < quality < 0.50 : 1300 elements Info : 0.50 < quality < 0.60 : 2319 elements Info : 0.60 < quality < 0.70 : 4434 elements Info : 0.70 < quality < 0.80 : 8026 elements Info : 0.80 < quality < 0.90 : 10853 elements Info : 0.90 < quality < 1.00 : 5615 elements Info : Done optimizing mesh (Wall 0.102774s, CPU 0.098523s) Info : 11797 nodes 75048 elements Info : Writing 'tests/canonical_model/mesh.msh'… Info : Done writing 'tests/canonical_model/mesh.msh' Info : Stopped on Fri Sep 12 13:16:26 2025 (From start: Wall 1.04062s, CPU 1.0539s) Created tests/canonical_model/params_nord1.txt Created tests/canonical_model/params_nord2.txt

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CI workflow completed at Fri Sep 12 13:16:38 UTC 2025